DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch : Electronics and Computer Engineering / Electronics and Computer Science Engineering

Semester :III Subject Code & Name: Computer Architecture & Operating System (BTESC304)

Max Marks: 60 Date:09-01-24 Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Draw the connections between the processor and main memory and explain the basic operational concepts.**	**CO1**	**6**
B)	**Explain the different kinds of addressing modes with an example.**	**CO3**	**6**
C)	**Explain the Intel X86 architecture with a neat block diagram.**	**CO1**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	**Explain briefly about Associate-mapped and set-associate mapped cache.**	**CO2**	**6**
B)	**What are the disadvantages of single continuous memory allocation? Explain.**	**CO2**	**6**
C)	**Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6 How many page faults would occur for the optimal page replacement algorithm, assuming three frames and all frames are initially empty.**	**CO4**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**Explain the basic concepts of instruction pipelining in computer architecture. How does instruction pipelining improve CPU performance?**	**CO3**	**6**
B)	**Draw the microinstruction-sequencing organization of next-address field and explain it.**	**CO3**	**6**
C)	**Explain input/output operations of computer architecture.**	**CO3**	**6**

Q.4	**Solve Any Two of the following.**		**12**
A)	**What is the difference between a process and a thread?**	**CO4**	**6**
B)	**Define the Operating System. Describe the operating system's objectives and functions.**	**CO4**	**6**
C)	**Assume the following workload in a system: Draw a Gantt chart illustrating the execution of these jobs using Round robin with time quantum = 2 unit scheduling algorithm and also Calculate the average waiting time and average turnaround time.**	**CO4**	**6**

Process	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

Q. 5	**Solve Any Two of the following.**		**12**
A)	**What is the critical section? What are the minimum requirements that should be satisfied by a solution to critical section problem?**	**CO5**	**6**
B)	**What is Producer Consumer problem? How it can illustrate the classical problem of synchronization? Explain.**	**CO5**	**6**
C)	**Explain deadlock avoidance using banker's algorithm with suitable example.**	**CO5**	**6**

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